

A Pannon Egyetem gyakorlatorientált infrastrukturális- és készségfejlesztési reformja

RRF-2.1.2.-21-2022-00007

NUMERICAL ANALYSIS

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Numerical Analysis

3 Linear Systems

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2025

3.1 Triangular Systems

Example

Solve the linear system

$$2x_1 - x_2 + 3x_3 + x_4 = 3$$

$$3x_2 - x_3 + 2x_4 = 13$$

$$2x_3 - x_4 = -2$$

$$3x_4 = 12$$

Solving the fourth equation for x_4 we get $x_4 = 4$. Substituting it to the third equation we get

$$x_3 = (-2 + x_4)/2 = 1.$$

Then the second equation yields

$$x_2 = (13 + x_3 - 2x_4)/3 = 2.$$

Finally, from the first equation we have

$$x_1 = (3 + x_2 - 3x_3 - x_4)/2 = -1.$$

We can generalize the method used in the previous example to solve the upper triangular n -dimensional linear system $\mathbf{Ax} = \mathbf{b}$, i.e., a linear system of the form

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\
 a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\
 &\vdots \\
 a_{nn}x_n &= b_n.
 \end{aligned} \tag{1}$$

We formulate the method of **backward substitution** in the following algorithm.

Algorithm: Backward substitution to solve a triangular system

INPUT: a_{ij} , $(i = 1, \dots, n, \quad j = 1, \dots, n)$, b_i , $(i = 1, \dots, n)$

OUTPUT: $x_1, \dots, x_n$

$$x_n \leftarrow b_n / a_{nn}$$

for $i = n - 1, \dots, 1$ **do**

$$x_i \leftarrow (b_i - \sum_{j=i+1}^n a_{ij}x_j) / a_{ii}$$

end do

output($x_1, x_2, \dots, x_n$)

The method of backward substitution can be performed if and only if

$$a_{ii} \neq 0 \quad \text{for all } i = 1, \dots, n.$$

Since

$$\det(\mathbf{A}) = a_{11}a_{22} \cdots a_{nn}$$

for a triangular matrix, the backward substitution works if and only if the system (1) has a unique solution, i.e.,

$$\det(\mathbf{A}) \neq 0.$$

The **time complexity** of the algorithm is

$$1 + 2 + \cdots + n = n(n+1)/2 = n^2/2 + \mathcal{O}(n)$$

multiplications and divisions, and

$$1 + 2 + \cdots + n - 1 = (n-1)n/2 = n^2/2 + \mathcal{O}(n)$$

additions and subtractions.

The notation $\mathcal{O}(n^k)$ denotes any polynomial with degree at most k .

3.2 Gaussian Elimination, Pivoting Strategies

Example

Consider the linear system

$$\begin{aligned}
 x_1 - 2x_2 - 2x_3 - 2x_4 &= -11 \\
 2x_1 - x_2 + 2x_3 + 4x_4 &= -8 \\
 -x_1 + 2x_2 + 3x_3 - 4x_4 &= 27 \\
 2x_1 + x_2 + 4x_3 - 2x_4 &= 28
 \end{aligned} \tag{2}$$

With the help of the first equation, the variable x_1 can be eliminated from the second, third and fourth equations. We multiply the first equation by 2, -1 and 2, respectively, and subtract it from the second, third and fourth equations, respectively:

$$\begin{aligned}
 x_1 - 2x_2 - 2x_3 - 2x_4 &= -11 \\
 3x_2 + 6x_3 + 8x_4 &= 14 \\
 x_3 - 6x_4 &= 16 \\
 -3x_2 \quad \quad - 6x_4 &= 6
 \end{aligned} \tag{3}$$

The resulting system is equivalent to (2).

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ \textcircled{2} & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \begin{matrix} \\ 2 \\ \\ \end{matrix} \sim$$

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ \textcircled{2} & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \begin{matrix} \\ 2 \\ \\ \end{matrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ & & & & \\ & & & & \end{pmatrix} \sim$$

$$2 - 2 \cdot 1 = 0, \quad -1 - 2 \cdot (-2) = 3, \quad 2 - 2 \cdot (-2) = 6,$$

$$4 - 2 \cdot (-2) = 8, \quad -8 - 2 \cdot (-11) = 14$$

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \begin{matrix} \\ \\ -1 \\ \end{matrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ & & & & \end{pmatrix} \sim$$

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ \textcircled{-1} & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \xrightarrow{-1} \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & 3 & 6 & 8 & 14 \end{pmatrix} \sim$$

$$\begin{aligned} -1 - (-1) \cdot 1 &= 0, & 2 - (-1) \cdot (-2) &= 0, & 3 - (-1) \cdot (-2) &= 1, \\ -4 - (-1) \cdot (-2) &= -6, & 27 - (-1) \cdot (-11) &= 16 \end{aligned}$$

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \xrightarrow{-2} \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \end{pmatrix} \sim$$

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ \textcircled{-2} & 1 & 4 & -2 & 28 \end{pmatrix} \xrightarrow{-2} \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \sim$$

$$\begin{aligned} -2 - (-2) \cdot 1 &= 0, & 1 - (-2) \cdot (-2) &= -3, & 4 - (-2) \cdot (-2) &= 0, \\ -2 - (-2) \cdot (-2) &= -6, & 28 - (-2) \cdot (-11) &= 6 \end{aligned}$$

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \stackrel{-1}{\sim}$$

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \stackrel{-1}{\sim} \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix}$$

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \sim$$

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix}^6$$

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \sim$$

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & 0 & 0 & 38 & -76 \end{pmatrix}$$

Example cont.

We associate the following 4×5 dimensional matrix to the system (2):

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

The elimination process is written shortly as

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \sim$$

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & 1 & -6 & 16 \\ 0 & 0 & 0 & 38 & -76 \end{pmatrix}$$

Using backward substitution we get: $x_1 = -3$, $x_2 = 2$, $x_3 = 4$, $x_4 = -2$.

Starting from the matrix $\tilde{\mathbf{A}}^{(0)}$ we obtain the sequence of matrices $\tilde{\mathbf{A}}^{(1)}$, $\tilde{\mathbf{A}}^{(2)}$, $\dots$, $\tilde{\mathbf{A}}^{(n-1)}$ describing equivalent linear systems in the following way. Let

$$\tilde{\mathbf{A}}^{(1)} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} & a_{1,n+1} \\ 0 & a_{22}^{(1)} & \cdots & a_{2n}^{(1)} & a_{2,n+1}^{(1)} \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & a_{n2}^{(1)} & \cdots & a_{nn}^{(1)} & a_{n,n+1}^{(1)} \end{pmatrix},$$

where

$$a_{ij}^{(1)} := a_{ij} - l_{i1}a_{1j}, \quad l_{i1} := \frac{a_{i1}}{a_{11}}, \quad i = 2, \dots, n, \quad j = 2, \dots, n+1,$$

(assuming $a_{11} \neq 0$).

If the matrices $\tilde{\mathbf{A}}^{(1)}, \dots, \tilde{\mathbf{A}}^{(k-1)}$ are defined for some $k \leq n-1$, then let

$$\tilde{\mathbf{A}}^{(k)} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1,k} & a_{1,k+1} & \cdots & a_{1,n} & a_{1,n+1} \\ 0 & a_{22}^{(1)} & \cdots & a_{2,k}^{(1)} & a_{2,k+1}^{(1)} & \cdots & a_{2,n}^{(1)} & a_{2,n+1}^{(1)} \\ & & \ddots & & & & & \\ 0 & 0 & \cdots & a_{k,k}^{(k-1)} & a_{k,k+1}^{(k-1)} & \cdots & a_{k,n}^{(k-1)} & a_{k,n+1}^{(k-1)} \\ 0 & 0 & \cdots & 0 & a_{k+1,k+1}^{(k)} & \cdots & a_{k+1,n}^{(k)} & a_{k+1,n+1}^{(k)} \\ \vdots & \vdots & & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & a_{n,k+1}^{(k)} & \cdots & a_{n,n}^{(k)} & a_{n,n+1}^{(k)} \end{pmatrix},$$

where

$$a_{ij}^{(k)} := a_{ij}^{(k-1)} - l_{ik} a_{kj}^{(k-1)}, \quad l_{ik} := \frac{a_{ik}^{(k-1)}}{a_{kk}^{(k-1)}}, \quad i = k+1, \dots, n, \quad j = k+1, \dots, n+1.$$

We perform these **elimination steps** for $k = 1, \dots, n-1$.

Finally, we solve the triangular system corresponding to the matrix $\tilde{\mathbf{A}}^{(n-1)}$ using the backward substitution method. The elements

$$a_{11}, a_{22}^{(1)}, \dots, a_{nn}^{(n-1)}$$

in the main diagonal of the last matrix of the Gaussian elimination are called **pivot elements**. Clearly, we can perform the Gaussian elimination if and only if all the pivot elements are nonzero.

If we perform the steps of the Gaussian elimination only on the coefficient matrix, the resulting matrices will be denoted by

$$\mathbf{A}^{(0)} := \mathbf{A}, \mathbf{A}^{(1)}, \dots, \mathbf{A}^{(n-1)}.$$

Algorithm: Gaussian elimination

INPUT: a_{ij} , ($i = 1, \dots, n$, $j = 1, \dots, n + 1$) - augmented matrix

OUTPUT: $x_1, \dots, x_n$

(elimination:)

```

for  $k = 1, \dots, n - 1$  do
    for  $i = k + 1, \dots, n$  do
         $l_{ik} \leftarrow a_{ik} / a_{kk}$ 
        for  $j = k + 1, \dots, n + 1$  do
             $a_{ij} \leftarrow a_{ij} - l_{ik} a_{kj}$ 
        end do
    end do
end do
  
```

Algorithm: Gaussian elimination (cont.)

(backward substitution:)

$$x_n \leftarrow a_{n,n+1}/a_{nn}$$

for $i = n - 1, \dots, 1$ **do**

$$x_i \leftarrow (a_{i,n+1} - \sum_{j=i+1}^n a_{ij}x_j)/a_{ii}$$

end do

output($x_1, x_2, \dots, x_n$)

The total number of multiplications and divisions needed for the elimination steps is

$$n^3/3 + n^2/2 - 5n/6,$$

and the number of additions and subtractions is

$$(n^3 - n)/3.$$

Together with the backward substitution,

$$n^3/3 + n^2/2 - 5n/6 + n^2/2 + n/2 = n^3/3 + n^2 - n/3 = n^3/3 + \mathcal{O}(n^2)$$

number of multiplications and divisions, and

$$(n^3 - n)/3 + n^2/2 - n/2 = n^3/3 + n^2/2 - 5n/6 = n^3/3 + \mathcal{O}(n^2)$$

number of additions and subtractions are needed to perform the Gaussian elimination. Shortly we say that the **time complexity** of the Gaussian elimination is

$$n^3/3$$

number of operations.

Example

Solve the system

$$\begin{aligned} 2x_1 - x_2 - 3x_4 &= 8 \\ 2x_1 - x_2 + x_3 + 5x_4 &= 2 \\ -3x_1 + x_2 + x_3 - 2x_4 &= -5 \\ 2x_1 + 4x_2 - x_4 &= 21 \end{aligned}$$

by Gaussian elimination. After performing the first step of the elimination we get

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 0 & 0 & 1 & 8 & -6 \\ 0 & -\frac{1}{2} & 1 & -\frac{13}{2} & 7 \\ 0 & 5 & 0 & 2 & 13 \end{pmatrix}.$$

Since the pivot element of the second row is 0, the Gaussian elimination cannot be continued.

On the other hand, the system has a unique solution: $x_1 = 4$, $x_2 = 3$, $x_3 = 2$ and $x_4 = -1$.

Example cont.

But if we change the second and third rows of the previous augmented matrix, the corresponding linear system is the same, and the elimination can be continued:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 0 & 0 & 1 & 8 & -6 \\ 0 & -\frac{1}{2} & 1 & -\frac{13}{2} & 7 \\ 0 & 5 & 0 & 2 & 13 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 0 & -\frac{1}{2} & 1 & -\frac{13}{2} & 7 \\ 0 & 0 & 1 & 8 & -6 \\ 0 & 5 & 0 & 2 & 13 \end{pmatrix} \sim$$

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 0 & -\frac{1}{2} & 1 & -\frac{13}{2} & 7 \\ 0 & 0 & 1 & 8 & -6 \\ 0 & 0 & 10 & -63 & 83 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 0 & -\frac{1}{2} & 1 & -\frac{13}{2} & 7 \\ 0 & 0 & 1 & 8 & -6 \\ 0 & 0 & 0 & -143 & 143 \end{pmatrix},$$

which yields the solution $x_1 = 4$, $x_2 = 3$, $x_3 = 2$ and $x_4 = -1$.

Example

Solve the linear system

$$0.0002x_1 - 30.5x_2 = -60.99$$

$$5.060x_1 - 1.05x_2 = 250.9$$

using Gaussian elimination and 4-digit arithmetic. First we compute the factor $l_{21} = 5.060/0.0002 = 25300$ (rounding to 4 significant digits). Then by multiplying the first equation by l_{21} and subtracting it from the second row we get

$$\begin{pmatrix} 0.0002 & -30.5 & -60.99 \\ 5.06 & -1.05 & 250.9 \end{pmatrix} \sim \begin{pmatrix} 0.0002 & -30.5 & -60.99 \\ 0 & 771700 & 1543000 \end{pmatrix}.$$

We note that we do not compute the first element of the second row by the Algorithm of the Gaussian elimination. Solving it we get the numerical solutions $\tilde{x}_1 = -100.0$ and $\tilde{x}_2 = 1.999$. We can check that the exact solution of the system is $x_1 = 50$ and $x_2 = 2$. Therefore, the relative errors of the numerical solutions are 300% and 0.05%, respectively.

Example cont.

Repeat the calculation for the system where we interchange the two equations:

$$\begin{pmatrix} 5.06 & -1.05 & 250.9 \\ 0.0002 & -30.5 & -60.99 \end{pmatrix} \sim \begin{pmatrix} 5.06 & -1.05 & 250.9 \\ 0 & -30.5 & -61.0 \end{pmatrix}.$$

This gives the numerical values $x_1 = 50.00$ and $x_2 = 2.000$, which are identical to the exact solutions.

What is the difference in between the two computations? In the first case, in order to compute l_{21} we needed to divide by a small number (0.0002), which gave us the increase of the rounding error. In the second case we performed the division by 5.06 in the computation of l_{21} , and we did not observe any error in the final result.

3.2 Gaussian Elimination, Pivoting Strategies

Partial Pivoting

Partial pivoting (or **maximal column pivoting**): before the k th step of the elimination, we select the element with the largest magnitude in the k th column in and under the main diagonal, i.e., let

$$|a_{lk}| = \max\{|a_{ik}| : i = k, \dots, n\}.$$

(The largest element is in the l th row.)

$$\begin{array}{c}
 \begin{array}{c} k \rightarrow \\ \updownarrow \\ l \rightarrow \end{array}
 \begin{pmatrix}
 a_{11} & a_{12} & \cdots & a_{1,k-1} & a_{1k} & a_{1,k+1} & \cdots & a_{1n} \\
 0 & a_{22} & \cdots & a_{2,k-1} & a_{2k} & a_{2,k+1} & \cdots & a_{2n} \\
 \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & & \vdots \\
 0 & 0 & \cdots & a_{k-1,k-1} & a_{k-1,k} & a_{k-1,k+1} & \cdots & a_{k-1,n} \\
 0 & 0 & \cdots & 0 & a_{kk} & a_{k,k+1} & \cdots & a_{kn} \\
 \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\
 0 & 0 & \cdots & 0 & a_{lk} & a_{l,k+1} & \cdots & a_{ln} \\
 \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\
 0 & 0 & \cdots & 0 & a_{nk} & a_{n,k+1} & \cdots & a_{nn}
 \end{pmatrix}
 \end{array}$$

We interchange the k th and l th rows, and then continue with the elimination.

Theorem:

The next statements are equivalent:

- (i) the linear system $\mathbf{Ax} = \mathbf{b}$ can be solved by Gaussian elimination with partial pivoting,*
- (ii) $\det(\mathbf{A}) \neq 0$,*
- (iii) the matrix $\mathbf{A}$ is invertible,*
- (iv) the linear system $\mathbf{Ax} = \mathbf{b}$ has a unique solution for all $\mathbf{b}$.*

Proof. We show that (i) and (ii) are equivalent.

Suppose first that (i) holds. Let $\mathbf{A}^{(0)} := \mathbf{A}$, and let $\mathbf{A}^{(k)}$ be the coefficient matrix in the Gaussian elimination after the k th step. The properties of the determinants yield that

$$\det(\mathbf{A}^{(k)}) = \det(\mathbf{A}^{(k-1)})$$

if there was no row change in the k th step, and

$$\det(\mathbf{A}^{(k)}) = -\det(\mathbf{A}^{(k-1)})$$

if there was a row change. Since the Gaussian elimination can be performed by the assumption, the triangular system corresponding to the coefficient matrix $\mathbf{A}^{(n-1)}$ of the last step is solvable, therefore

$$\det(\mathbf{A}^{(n-1)}) \neq 0.$$

But this implies

$$\det(\mathbf{A}) = \pm \det(\mathbf{A}^{(n-1)}) \neq 0.$$

Suppose now that the Gaussian elimination with partial pivoting terminates before the k th step. Then $a_{ik}^{(k-1)} = 0$ for all $i = k, \dots, n$, i.e.,

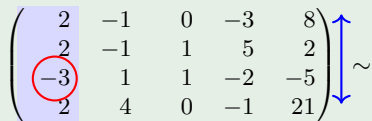
$$\mathbf{A}^{(k-1)} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1,k-1} & a_{1k} & a_{k,k+1} & \cdots & a_{1n} \\ 0 & a_{22}^{(1)} & \cdots & a_{2,k-1}^{(1)} & a_{2k}^{(1)} & a_{2,k+1}^{(1)} & \cdots & a_{2n}^{(1)} \\ & & \ddots & & & & & \\ 0 & 0 & \cdots & a_{k-1,k-1}^{(k-2)} & a_{k-1,k}^{(k-2)} & a_{k-1,k+1}^{(k-2)} & \cdots & a_{k-1,n}^{(k-2)} \\ 0 & 0 & \cdots & 0 & 0 & a_{k,k+1}^{(k-1)} & \cdots & a_{kn}^{(k-1)} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 & 0 & a_{n,k+1}^{(k-1)} & \cdots & a_{nn}^{(k-1)} \end{pmatrix}.$$

Hence

$$\begin{aligned} \det(\mathbf{A}^{(k-1)}) &= a_{11} a_{22}^{(1)} \cdots a_{k-1,k-1}^{(k-2)} \det \begin{pmatrix} 0 & a_{k,k+1}^{(k-1)} & \cdots & a_{kn}^{(k-1)} \\ \vdots & \vdots & & \vdots \\ 0 & a_{n,k+1}^{(k-1)} & \cdots & a_{nn}^{(k-1)} \end{pmatrix} \\ &= 0 = \det(\mathbf{A}). \end{aligned}$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\left(\begin{array}{ccccc} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{array} \right) \sim$$


Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\left(\begin{array}{ccccc} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{array} \right) \begin{array}{c} \uparrow \\ \sim \\ \downarrow \end{array} \left(\begin{array}{ccccc} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{array} \right) \sim$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ \boxed{2} & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \stackrel{-\frac{2}{3}}{\sim}$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ \boxed{2} & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \begin{matrix} \\ -\frac{2}{3} \\ \\ \end{matrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \end{pmatrix} \sim$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \begin{matrix} -\frac{2}{3} \\ \\ \\ \end{matrix} \sim \\
 \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \end{pmatrix} \sim$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \end{pmatrix}$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \begin{matrix} -\frac{2}{3} \\ \\ \\ \end{matrix} \sim \\
 \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \end{pmatrix} \sim$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \xrightarrow{-\frac{2}{3}} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix}$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim$$

$$\begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \sim$$

The matrix is shown with a blue vertical double-headed arrow indicating row operations between the second and third rows. The element $\frac{14}{3}$ in the third row, second column is circled in red.

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim$$

$$\begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \begin{matrix} \updownarrow \\ \sim \end{matrix} \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \end{pmatrix} \sim$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{aligned}
 & \begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \end{pmatrix} \xrightarrow{-\frac{1}{14}} \sim
 \end{aligned}$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{aligned}
 & \begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \end{pmatrix} \xrightarrow{-\frac{1}{14}} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{2} & \frac{83}{14} \end{pmatrix} \sim
 \end{aligned}$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \end{pmatrix} \xrightarrow{-\frac{1}{14}} \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{2} & \frac{83}{14} \end{pmatrix}$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{aligned}
 & \begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \end{pmatrix} \xrightarrow{-\frac{1}{14}} \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{2} & \frac{83}{14} \\ 0 & 0 & \frac{12}{7} & \frac{7}{2} & -\frac{1}{14} \end{pmatrix}
 \end{aligned}$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{2} & \frac{83}{14} \\ 0 & 0 & \frac{12}{7} & \frac{7}{2} & -\frac{1}{14} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{2} & \frac{83}{14} \\ 0 & 0 & \frac{12}{7} & \frac{7}{2} & -\frac{1}{14} \end{pmatrix}$$

A blue double-headed vertical arrow indicates the row swap between the third and fourth rows of the final matrix. The element $\frac{12}{7}$ in the fourth row, third column is circled in red.

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{aligned}
 & \begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{7} & \frac{83}{7} \\ 0 & 0 & \frac{12}{7} & -\frac{1}{7} & -\frac{14}{14} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{12}{7} & -\frac{1}{7} & -\frac{14}{14} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{7} & \frac{83}{7} \end{pmatrix} \sim
 \end{aligned}$$

A blue double-headed vertical arrow indicates a row swap between the third and fourth rows of the final matrix. The element $\frac{12}{7}$ in the third row, third column is circled in red.

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim$$

$$\begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \end{pmatrix} \sim$$

$$\begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{2} & \frac{83}{14} \\ 0 & 0 & \frac{12}{7} & \frac{7}{2} & -\frac{1}{14} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{12}{7} & \frac{7}{2} & -\frac{1}{14} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{2} & \frac{83}{14} \end{pmatrix} \sim$$

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{aligned}
 & \begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{7} & \frac{83}{7} \\ 0 & 0 & \frac{12}{7} & -\frac{7}{2} & -\frac{1}{14} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{12}{7} & -\frac{7}{2} & -\frac{1}{14} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{7} & \frac{83}{7} \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{12}{7} & -\frac{7}{2} & -\frac{1}{14} \\ 0 & 0 & 0 & -\frac{143}{24} & \frac{143}{24} \end{pmatrix}
 \end{aligned}$$

In the final matrix, the element $\frac{5}{7}$ in the third row, fourth column is circled in blue, and the value $\frac{5}{12}$ is written in blue next to the final matrix.

Example

Solve the previous system using Gaussian elimination with partial pivoting:

$$\begin{aligned}
 & \begin{pmatrix} 2 & -1 & 0 & -3 & 8 \\ 2 & -1 & 1 & 5 & 2 \\ -3 & 1 & 1 & -2 & -5 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 2 & -1 & 1 & 5 & 2 \\ 2 & -1 & 0 & -3 & 8 \\ 2 & 4 & 0 & -1 & 21 \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{13}{3} & \frac{14}{3} \\ 0 & -\frac{1}{3} & \frac{5}{3} & \frac{11}{3} & -\frac{4}{3} \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{7} & \frac{83}{7} \\ 0 & 0 & \frac{12}{7} & \frac{2}{7} & -\frac{1}{14} \end{pmatrix} \sim \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{12}{7} & \frac{2}{7} & -\frac{1}{14} \\ 0 & 0 & \frac{5}{7} & -\frac{9}{7} & \frac{83}{7} \end{pmatrix} \sim \\
 & \begin{pmatrix} -3 & 1 & 1 & -2 & -5 \\ 0 & \frac{14}{3} & \frac{2}{3} & -\frac{7}{3} & \frac{53}{3} \\ 0 & 0 & \frac{12}{7} & \frac{2}{7} & -\frac{1}{14} \\ 0 & 0 & 0 & -\frac{143}{24} & \frac{143}{24} \end{pmatrix}
 \end{aligned}$$

The solution is $x_1 = 4$, $x_2 = 3$, $x_3 = 2$ and $x_4 = -1$.

3.2 Gaussian Elimination, Pivoting Strategies

Complete Pivoting

complete pivoting or **maximal pivoting**: before the k th step of the elimination we find the first row index l and column index m such that

$$|a_{lm}| = \max\{|a_{ij}| : i = k, \dots, n, j = k, \dots, n\}.$$

(The largest element is located in the l th row and in the m th column.)

$$\begin{array}{c}
 \begin{array}{c} k \rightarrow \\ \updownarrow \\ l \rightarrow \end{array}
 \begin{pmatrix}
 a_{11} & a_{12} & \cdots & a_{1,k-1} & a_{1k} & \cdots & a_{1m} & \cdots & a_{1n} & a_{1,n+1} \\
 0 & a_{22} & \cdots & a_{2,k-1} & a_{2k} & \cdots & a_{2m} & \cdots & a_{2n} & a_{2,n+1} \\
 \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots & & \vdots & \vdots \\
 0 & 0 & \cdots & a_{k-1,k-1} & a_{k-1,k} & \cdots & a_{k-1,m} & \cdots & a_{k-1,n} & a_{k-1,n+1} \\
 0 & 0 & \cdots & 0 & a_{kk} & \cdots & a_{km} & \cdots & a_{kn} & a_{k,n+1} \\
 \vdots & \vdots & & \vdots & \vdots & & \vdots & & \vdots & \vdots \\
 0 & 0 & \cdots & 0 & a_{lk} & \cdots & a_{lm} & \cdots & a_{ln} & a_{l,n+1} \\
 \vdots & \vdots & & \vdots & \vdots & & \vdots & & \vdots & \vdots \\
 0 & 0 & \cdots & 0 & a_{nk} & \cdots & a_{nm} & \cdots & a_{nn} & a_{n,n+1}
 \end{pmatrix}
 \end{array}$$

Then we interchange the k th and l th rows and the k th and m th columns, and we continue with the elimination step.

Example

Consider again the system examined in earlier examples, and here we solve it using Gaussian elimination with complete pivoting:

$$\left(\begin{array}{cccc|c} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{array} \right) \sim$$

x_1 x_2 x_3 x_4

Example

Consider again the system examined in earlier examples, and here we solve it using Gaussian elimination with complete pivoting:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 2 & 4 & -8 \\ 1 & -2 & -2 & -2 & -11 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

$x_1 \quad x_2 \quad x_3 \quad x_4$

Example

Consider again the system examined in earlier examples, and here we solve it using Gaussian elimination with complete pivoting:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 2 & 4 & -8 \\ 1 & -2 & -2 & -2 & -11 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ -2 & -2 & -2 & 1 & -11 \\ -4 & 2 & 3 & -1 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix}$$

$x_1 \quad x_2 \quad x_3 \quad x_4$
 $x_1 \quad x_2 \quad x_3 \quad x_4$
 $x_4 \quad x_2 \quad x_3 \quad x_1$

Example

Consider again the system examined in earlier examples, and here we solve it using Gaussian elimination with complete pivoting:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 2 & 4 & -8 \\ 1 & -2 & -2 & -2 & -11 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim$$

$\begin{matrix} x_1 & x_2 & x_3 & x_4 \end{matrix}$

$$\begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ -2 & -2 & -2 & 1 & -11 \\ -4 & 2 & 3 & -1 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim$$

$\begin{matrix} x_4 & x_2 & x_3 & x_1 \end{matrix}$

Example

Consider again the system examined in earlier examples, and here we solve it using Gaussian elimination with complete pivoting:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 2 & 4 & -8 \\ 1 & -2 & -2 & -2 & -11 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim$$

x_1 x_2 x_3 x_4

$$\begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ -2 & -2 & -2 & 1 & -11 \\ -4 & 2 & 3 & -1 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ 0 & -\frac{5}{2} & -1 & 2 & -15 \\ 0 & 1 & 5 & 1 & 19 \\ 0 & \frac{1}{2} & 5 & -1 & 24 \end{pmatrix} \sim$$

x_4 x_2 x_3 x_1

Example

Consider again the system examined in earlier examples, and here we solve it using Gaussian elimination with complete pivoting:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 2 & 4 & -8 \\ 1 & -2 & -2 & -2 & -11 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim$$

$x_1 \quad x_2 \quad x_3 \quad x_4$ $x_1 \quad x_2 \quad x_3 \quad x_4$

$$\begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ -2 & -2 & -2 & 1 & -11 \\ -4 & 2 & 3 & -1 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ 0 & -\frac{5}{2} & -1 & 2 & -15 \\ 0 & 1 & 5 & 1 & 19 \\ 0 & \frac{1}{2} & 5 & -1 & 24 \end{pmatrix} \sim$$

$x_4 \quad x_2 \quad x_3 \quad x_1$ $x_4 \quad x_2 \quad x_3 \quad x_1$

Example

Consider again the system examined in earlier examples, and here we solve it using Gaussian elimination with complete pivoting:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 2 & 4 & -8 \\ 1 & -2 & -2 & -2 & -11 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim$$

$x_1 \quad x_2 \quad x_3 \quad x_4$ $x_1 \quad x_2 \quad x_3 \quad x_4$

$$\begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ -2 & -2 & -2 & 1 & -11 \\ -4 & 2 & 3 & -1 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ 0 & -\frac{5}{2} & -1 & 2 & -15 \\ 0 & 1 & 5 & 1 & 19 \\ 0 & \frac{1}{2} & 5 & -1 & 24 \end{pmatrix} \sim$$

$x_4 \quad x_2 \quad x_3 \quad x_1$ $x_4 \quad x_2 \quad x_3 \quad x_1$

$$\begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ 0 & 1 & 5 & 1 & 19 \\ 0 & -\frac{5}{2} & -1 & 2 & -15 \\ 0 & \frac{1}{2} & 5 & -1 & 24 \end{pmatrix} \sim$$

$x_4 \quad x_2 \quad x_3 \quad x_1$

Example

Consider again the system examined in earlier examples, and here we solve it using Gaussian elimination with complete pivoting:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 2 & -1 & 2 & 4 & -8 \\ 1 & -2 & -2 & -2 & -11 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim$$

$x_1 \quad x_2 \quad x_3 \quad x_4$ $x_1 \quad x_2 \quad x_3 \quad x_4$

$$\begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ -2 & -2 & -2 & 1 & -11 \\ -4 & 2 & 3 & -1 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ 0 & -\frac{5}{2} & -1 & 2 & -15 \\ 0 & 1 & 5 & 1 & 19 \\ 0 & \frac{1}{2} & 5 & -1 & 24 \end{pmatrix} \sim$$

$x_4 \quad x_2 \quad x_3 \quad x_1$ $x_4 \quad x_2 \quad x_3 \quad x_1$

$$\begin{pmatrix} 4 & -1 & 2 & 2 & -8 \\ 0 & 1 & 5 & 1 & 19 \\ 0 & -\frac{5}{2} & -1 & 2 & -15 \\ 0 & \frac{1}{2} & 5 & -1 & 24 \end{pmatrix} \sim \begin{pmatrix} 4 & 2 & -1 & 2 & -8 \\ 0 & 5 & 1 & 1 & 19 \\ 0 & -1 & -\frac{5}{2} & 2 & -15 \\ 0 & 5 & \frac{1}{2} & -1 & 24 \end{pmatrix} \sim$$

$x_4 \quad x_2 \quad x_3 \quad x_1$ $x_4 \quad x_3 \quad x_2 \quad x_1$

Example cont.

$$\begin{pmatrix}
 4 & 2 & -1 & 2 & -8 \\
 0 & 5 & 1 & 1 & 19 \\
 0 & -1 & -\frac{5}{2} & 2 & -15 \\
 0 & 5 & \frac{1}{2} & -1 & 24
 \end{pmatrix} \sim$$

x_4 x_3 x_2 x_1

Example cont.

$$\begin{pmatrix}
 4 & 2 & -1 & 2 & -8 \\
 0 & 5 & 1 & 1 & 19 \\
 0 & -1 & -\frac{5}{2} & 2 & -15 \\
 0 & 5 & \frac{1}{2} & -1 & 24
 \end{pmatrix} \sim \begin{pmatrix}
 4 & 2 & -1 & 2 & -8 \\
 0 & 5 & 1 & 1 & 19 \\
 0 & 0 & -\frac{23}{10} & \frac{11}{5} & -\frac{56}{5} \\
 0 & 0 & -\frac{1}{2} & -2 & 5
 \end{pmatrix} \sim$$

x_4 x_3 x_2 x_1

Example cont.

$$\begin{pmatrix}
 4 & 2 & -1 & 2 & -8 \\
 0 & 5 & 1 & 1 & 19 \\
 0 & -1 & -\frac{5}{2} & 2 & -15 \\
 0 & 5 & \frac{1}{2} & -1 & 24
 \end{pmatrix} \sim \begin{pmatrix}
 4 & 2 & -1 & 2 & -8 \\
 0 & 5 & 1 & 1 & 19 \\
 0 & 0 & -\frac{23}{10} & \frac{11}{5} & -\frac{56}{5} \\
 0 & 0 & -\frac{1}{2} & -2 & 5
 \end{pmatrix} \sim$$

x_4 x_3 x_2 x_1

Example cont.

$$\begin{pmatrix}
 4 & 2 & -1 & 2 & -8 \\
 0 & 5 & 1 & 1 & 19 \\
 0 & -1 & -\frac{5}{2} & 2 & -15 \\
 0 & 5 & \frac{1}{2} & -1 & 24
 \end{pmatrix} \sim \begin{pmatrix}
 4 & 2 & -1 & 2 & -8 \\
 0 & 5 & 1 & 1 & 19 \\
 0 & 0 & -\frac{23}{10} & \frac{11}{5} & -\frac{56}{5} \\
 0 & 0 & -\frac{1}{2} & -2 & 5
 \end{pmatrix} \sim$$

x_4 x_3 x_2 x_1

Example cont.

$$\begin{pmatrix}
 4 & 2 & -1 & 2 & -8 \\
 0 & 5 & 1 & 1 & 19 \\
 0 & -1 & -\frac{5}{2} & 2 & -15 \\
 0 & 5 & \frac{1}{2} & -1 & 24
 \end{pmatrix}
 \underset{\substack{x_4 \quad x_3 \quad x_2 \quad x_1}}{\sim}
 \begin{pmatrix}
 4 & 2 & -1 & 2 & -8 \\
 0 & 5 & 1 & 1 & 19 \\
 0 & 0 & -\frac{23}{10} & \frac{11}{5} & -\frac{56}{5} \\
 0 & 0 & -\frac{1}{2} & -2 & 5
 \end{pmatrix}
 \underset{\substack{x_4 \quad x_3 \quad x_2 \quad x_1}}{\sim}
 \begin{pmatrix}
 4 & 2 & -1 & 2 & -8 \\
 0 & 5 & 1 & 1 & 19 \\
 0 & 0 & -\frac{23}{10} & \frac{11}{5} & -\frac{56}{5} \\
 0 & 0 & 0 & -\frac{57}{23} & \frac{171}{23}
 \end{pmatrix}
 \underset{\substack{x_4 \quad x_3 \quad x_2 \quad x_1}}{}$$

The corresponding triangular system is

$$\begin{aligned}
 4x_4 + 2x_3 - x_2 + 2x_1 &= -8 \\
 5x_3 + x_2 + x_1 &= 19 \\
 -\frac{23}{10}x_2 + \frac{11}{5}x_1 &= -\frac{56}{5} \\
 -\frac{57}{23}x_1 &= \frac{171}{23}
 \end{aligned}$$

The result is again $x_1 = -3$, $x_2 = 2$, $x_3 = 4$ and $x_4 = -2$.

A square matrix $A \in \mathbb{R}^{n \times n}$ is called **row diagonally dominant** or simply **diagonally dominant** if

$$|a_{ii}| > \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}|, \quad i = 1, \dots, n.$$

Example

The matrix

$$\begin{pmatrix} 5 & -1 & 2 \\ 2 & -10 & 1 \\ -2 & 0 & 3 \end{pmatrix}$$

is diagonally dominant, since

$$5 > 1 + 2$$

$$10 > 2 + 1$$

$$3 > 2 + 0.$$

Theorem:

If $\mathbf{A} \in \mathbb{R}^{n \times n}$ is diagonally dominant, then $\mathbf{A}$ is invertible.

Proof

Suppose that $\mathbf{A}$ is not invertible. Then the linear system $\mathbf{A}\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\mathbf{x} \neq \mathbf{0}$. Let k be such that

$$|x_k| = \max\{|x_i| : i = 1, \dots, n\}.$$

Then $x_k \neq 0$. Since $\sum_{j=1}^n a_{ij}x_j = 0$ for all $i = 1, \dots, n$, we get $a_{kk}x_k = -\sum_{j=1, j \neq k}^n a_{kj}x_j$. Then the triangle-inequality yields

$$|a_{kk}x_k| \leq \sum_{\substack{j=1 \\ j \neq k}}^n |a_{kj}x_j|,$$

and so

$$|a_{kk}| \leq \sum_{\substack{j=1 \\ j \neq k}}^n |a_{kj}| \frac{|x_j|}{|x_k|} \leq \sum_{\substack{j=1 \\ j \neq k}}^n |a_{kj}|,$$

which is a contradiction.

Theorem:

If the matrix $\mathbf{A}$ is diagonally dominant, then the Gaussian elimination can be performed on the linear system $\mathbf{Ax} = \mathbf{b}$ without pivoting, and the method is stable with respect to the rounding errors.

The square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ is called **positive definite** (**negative definite**) if $\mathbf{A}$ is symmetric and

$$\mathbf{x}^T \mathbf{A} \mathbf{x} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j > 0, \quad \mathbf{x} \neq \mathbf{0},$$

($\mathbf{x}^T \mathbf{A} \mathbf{x} < 0$, respectively, for all $\mathbf{x} \neq \mathbf{0}$). The matrix $\mathbf{A}$ is called **positive semi-definite** (**negative semi-definite**) if $\mathbf{A}$ is symmetric and $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$ ($\mathbf{x}^T \mathbf{A} \mathbf{x} \leq 0$, respectively) for all $\mathbf{x}$.

Example

The matrix

$$\mathbf{A} = \begin{pmatrix} 4 & -1 \\ -1 & 2 \end{pmatrix}$$

is positive definite since it is symmetric, and for $\mathbf{x} \neq \mathbf{0}$

$$\begin{aligned} \mathbf{x}^T \mathbf{A} \mathbf{x} &= (x_1 \ x_2) \begin{pmatrix} 4 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (x_1 \ x_2) \begin{pmatrix} 4x_1 - x_2 \\ -x_1 + 2x_2 \end{pmatrix} \\ &= 4x_1^2 - 2x_1x_2 + 2x_2^2 = 3x_1^2 + (x_1 - x_2)^2 + x_2^2 > 0. \end{aligned}$$

Theorem:

The symmetric matrix $\mathbf{A}$ is positive definite if and only if all of its upper left minors, the so-called principal minors are positive, i.e.,

$$\det \begin{pmatrix} a_{11} & \cdots & a_{1i} \\ \vdots & & \vdots \\ a_{i1} & \cdots & a_{ii} \end{pmatrix} > 0, \quad i = 1, 2, \dots, n.$$

$$\left(\begin{array}{c|c|c|c|c} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ \hline a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ \hline a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \\ \hline a_{41} & a_{42} & a_{43} & a_{44} & a_{45} \\ \hline a_{51} & a_{52} & a_{53} & a_{54} & a_{55} \end{array} \right)$$

Theorem:

Let $\mathbf{A}$ be a symmetric $n \times n$ matrix, $\mathbf{b} \in \mathbb{R}^n$. Then $\mathbf{A}$ is positive definite if and only if the Gaussian elimination can be performed for the system $\mathbf{Ax} = \mathbf{b}$ without pivoting, and the pivot elements are all positive. Moreover, the method is stable with respect to the rounding errors.

3.3 Gauss–Jordan Elimination

A version of the Gaussian elimination is the **Gauss–Jordan elimination**, where we use the elimination steps of the Gaussian elimination to transform the coefficient matrix part of the augmented matrix to the identity matrix, i.e., the matrix

$$(\mathbf{A}, \mathbf{b})$$

is converted to the form

$$(\mathbf{I}, \mathbf{b}^{(n-1)}).$$

Then the solution of the linear system $\mathbf{Ax} = \mathbf{b}$ is

$$\mathbf{x} = \mathbf{b}^{(n-1)}.$$

Algorithm: Gauss–Jordan elimination

INPUT: a_{ij} , ($i = 1, \dots, n$, $j = 1, \dots, n + 1$)

OUTPUT: $x_1, \dots, x_n$

(converting the coefficients to a diagonal form:)

for $k = 1, \dots, n$ **do**

for $i = 1, \dots, n$ **do**

if $i \neq k$ **do**

$l_{ik} \leftarrow a_{ik}/a_{kk}$

for $j = k + 1, \dots, n + 1$ **do**

$a_{ij} \leftarrow a_{ij} - l_{ik}a_{kj}$

end do

end do

end do

end do

for $i = 1, \dots, n$ **do**

$x_i \leftarrow a_{i,n+1}/a_{ii}$

end do

output($x_1, x_2, \dots, x_n$)

It can be checked that the **time complexity** of the Gauss-Jordan elimination is

$$n^3/2 + \mathcal{O}(n^2)$$

number of multiplications and divisions.

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\left(\begin{array}{ccccc} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ -2 & 1 & 4 & -2 & 28 \end{array} \right)^2 \sim$$

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\left(\begin{array}{ccccc} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ -2 & 1 & 4 & -2 & 28 \end{array} \right)^2 \sim \left(\begin{array}{ccccc} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \end{array} \right) \sim$$

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\left(\begin{array}{ccccc} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ -2 & 1 & 4 & -2 & 28 \end{array} \right) \begin{array}{c} \\ \\ -1 \\ \end{array} \sim \left(\begin{array}{ccccc} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \end{array} \right) \sim$$

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\left(\begin{array}{ccccc} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ \textcircled{-2} & 1 & 4 & -2 & 28 \end{array} \right) \xrightarrow{-2} \left(\begin{array}{ccccc} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & -3 & 0 & -6 & 6 \end{array} \right) \sim$$

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & \boxed{-2} & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \xrightarrow{-\frac{3}{2}} \sim \begin{pmatrix} 1 & 0 & 2 & \frac{10}{3} & -\frac{5}{3} \\ 0 & 3 & 6 & 8 & 14 \end{pmatrix} \sim$$

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \stackrel{-1}{\sim} \begin{pmatrix} 1 & 0 & 2 & \frac{10}{3} & -\frac{5}{3} \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix}$$

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \sim \\
 \begin{pmatrix} 1 & 0 & \boxed{2} & \frac{10}{3} & -\frac{5}{3} \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix} \xrightarrow{-\frac{2}{5}} \begin{pmatrix} 1 & 0 & 0 & \frac{14}{15} & -\frac{73}{15} \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix} \sim$$

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \sim \\
 \begin{pmatrix} 1 & 0 & 2 & \frac{10}{3} & -\frac{5}{3} \\ 0 & 3 & \textcircled{6} & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix} \xrightarrow{-\frac{6}{5}} \begin{pmatrix} 1 & 0 & 0 & \frac{14}{15} & -\frac{73}{15} \\ 0 & 3 & 0 & \frac{4}{5} & \frac{22}{5} \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & -5 & -6 & -8 \end{pmatrix} \sim$$

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \sim \\
 \begin{pmatrix} 1 & 0 & 2 & \frac{10}{3} & -\frac{5}{3} \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & \textcircled{6} & 2 & 20 \end{pmatrix} \xrightarrow{-\frac{6}{5}} \begin{pmatrix} 1 & 0 & 0 & \frac{14}{15} & -\frac{73}{15} \\ 0 & 3 & 0 & \frac{4}{5} & \frac{22}{5} \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & 0 & -\frac{26}{5} & \frac{52}{5} \end{pmatrix} \sim$$

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \sim \\
 \begin{pmatrix} 1 & 0 & 2 & \frac{10}{3} & -\frac{5}{3} \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & \frac{14}{15} & -\frac{73}{15} \\ 0 & 3 & 0 & \frac{4}{5} & \frac{22}{5} \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & 0 & -\frac{26}{5} & \frac{52}{5} \end{pmatrix} \sim \\
 \begin{pmatrix} 1 & 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & -\frac{26}{5} & \frac{52}{5} \end{pmatrix} \sim$$

Example

We apply the Gauss–Jordan elimination to the system examined before:

$$\begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & -3 & -4 & 3 \\ -2 & 1 & 4 & -2 & 28 \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & -2 & -2 & -11 \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & -3 & 0 & -6 & 6 \end{pmatrix} \sim \\
 \begin{pmatrix} 1 & 0 & 2 & \frac{10}{3} & -\frac{5}{3} \\ 0 & 3 & 6 & 8 & 14 \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & \frac{14}{15} & -\frac{73}{15} \\ 0 & 3 & 0 & \frac{4}{5} & \frac{22}{5} \\ 0 & 0 & -5 & -6 & -8 \\ 0 & 0 & 0 & -\frac{26}{5} & \frac{52}{5} \end{pmatrix} \sim -\frac{2}{13} \\
 \begin{pmatrix} 1 & 0 & 0 & 0 & -3 \\ 0 & 3 & 0 & 0 & 6 \\ 0 & 0 & 0 & -\frac{26}{5} & \frac{52}{5} \end{pmatrix} \sim$$

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 \begin{pmatrix} 1 & 0 & 0 & 0 & -3 \\ 0 & 3 & 0 & 0 & 6 \\ 0 & 0 & -5 & 0 & -20 \\ 0 & 0 & 0 & -\frac{26}{5} & \frac{52}{5} \end{pmatrix} \sim$$

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 \begin{pmatrix} 1 & 0 & 0 & 0 & -3 \\ 0 & 3 & 0 & 0 & 6 \\ 0 & 0 & -5 & 0 & -20 \\ 0 & 0 & 0 & -\frac{26}{5} & \frac{52}{5} \end{pmatrix} \begin{matrix} \\ \frac{1}{3} \\ -\frac{1}{5} \\ -\frac{5}{26} \end{matrix} \sim$$

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The last column gives $x_1 = -3$, $x_2 = 2$, $x_3 = 4$ and $x_4 = -2$.

We can combine pivoting strategies together with the Gauss–Jordan elimination.

Example

Here we apply the Gauss–Jordan elimination with partial pivoting:

$$\left(\begin{array}{ccccc} 1 & -2 & -2 & -2 & -11 \\ 2 & -1 & 2 & 4 & -8 \\ -1 & 2 & 3 & -4 & 27 \\ -2 & 1 & 4 & -2 & 28 \end{array} \right) \begin{array}{c} \updownarrow \\ \sim \end{array}$$

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 \begin{pmatrix} 2 & -1 & 2 & 4 & -8 \\ 0 & -\frac{3}{2} & -3 & -4 & -7 \\ 0 & \frac{3}{2} & 4 & -2 & 23 \\ 0 & 0 & 6 & 2 & 20 \end{pmatrix} \sim$$

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Example cont.

$$\left(\begin{array}{ccccc} 2 & 0 & 4 & \frac{20}{3} & -\frac{10}{3} \\ 0 & -\frac{3}{2} & -3 & -4 & -7 \\ 0 & 0 & 6 & 2 & 20 \\ 0 & 0 & 1 & -6 & 16 \end{array} \right) \sim$$

Example cont.

$$\begin{pmatrix} 2 & 0 & 4 & \frac{20}{3} & -\frac{10}{3} \\ 0 & -\frac{3}{2} & -3 & -4 & -7 \\ 0 & 0 & 6 & 2 & 20 \\ 0 & 0 & 1 & -6 & 16 \end{pmatrix} \sim \begin{pmatrix} 2 & 0 & 0 & \frac{16}{3} & -\frac{50}{3} \\ 0 & -\frac{3}{2} & 0 & -3 & 3 \\ 0 & 0 & 6 & 2 & 20 \\ 0 & 0 & 0 & -\frac{19}{3} & \frac{38}{3} \end{pmatrix} \sim \\
 \begin{pmatrix} 2 & 0 & 0 & 0 & -6 \\ 0 & -\frac{3}{2} & 0 & 0 & -3 \\ 0 & 0 & 6 & 0 & 24 \\ 0 & 0 & 0 & -\frac{19}{3} & \frac{38}{3} \end{pmatrix}$$

Example cont.

$$\begin{pmatrix} 2 & 0 & 4 & \frac{20}{3} & -\frac{10}{3} \\ 0 & -\frac{3}{2} & -3 & -4 & -7 \\ 0 & 0 & 6 & 2 & 20 \\ 0 & 0 & 1 & -6 & 16 \end{pmatrix} \sim \begin{pmatrix} 2 & 0 & 0 & \frac{16}{3} & -\frac{50}{3} \\ 0 & -\frac{3}{2} & 0 & -3 & 3 \\ 0 & 0 & 6 & 2 & 20 \\ 0 & 0 & 0 & -\frac{19}{3} & \frac{38}{3} \end{pmatrix} \sim$$

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Example cont.

$$\begin{pmatrix} 2 & 0 & 4 & \frac{20}{3} & -\frac{10}{3} \\ 0 & -\frac{3}{2} & -3 & -4 & -7 \\ 0 & 0 & 6 & 2 & 20 \\ 0 & 0 & 1 & -6 & 16 \end{pmatrix} \sim \begin{pmatrix} 2 & 0 & 0 & \frac{16}{3} & -\frac{50}{3} \\ 0 & -\frac{3}{2} & 0 & -3 & 3 \\ 0 & 0 & 6 & 2 & 20 \\ 0 & 0 & 0 & -\frac{19}{3} & \frac{38}{3} \end{pmatrix} \sim \\
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Therefore, the solution is $x_1 = -3$, $x_2 = 2$, $x_3 = 4$ and $x_4 = -2$.

3.4 Tridiagonal Linear Systems

We say that a square matrix (a_{ij}) is **tridiagonal** if

$$a_{ij} = 0 \quad \text{for all } |i - j| > 1,$$

i.e., nonzero numbers can appear only in the main diagonal and in the next diagonal above and under it.

We will use the following notations:

$$\begin{pmatrix} d_1 & c_1 & 0 & 0 & \cdots & 0 \\ a_1 & d_2 & c_2 & 0 & \cdots & 0 \\ 0 & a_2 & d_3 & c_3 & \cdots & 0 \\ & & \ddots & \ddots & \ddots & \\ 0 & 0 & \cdots & a_{n-2} & d_{n-1} & c_{n-1} \\ 0 & 0 & \cdots & 0 & a_{n-1} & d_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_{n-1} \\ b_n \end{pmatrix}. \quad (5)$$

It is practical to store the elements of a tridiagonal matrix in three vectors (a_i) , (d_i) and (c_i) , as it is used above. In this case only

$$3n - 2$$

storage area is needed for the coefficients.

It is clear that applying the Gaussian elimination to the system (5) the elements a_i below the main diagonal will become 0, and the numbers c_i will not be changed during the elimination steps. We have to compute the new values of the variables d_i and b_i during the elimination. In the next algorithm we override the old values of the vectors (d_i) and (b_i) with the actual new ones.

Algorithm: Gaussian elimination for tridiagonal linear systems

INPUT: a_i, c_i ($i = 1, \dots, n-1$), d_i, b_i ($i = 1, \dots, n$)

OUTPUT: $x_1, \dots, x_n$

(elimination:)

for $i = 2, \dots, n$ **do**

$temp \leftarrow a_{i-1}/d_{i-1}$

$d_i \leftarrow d_i - temp \cdot c_{i-1}$

$b_i \leftarrow b_i - temp \cdot b_{i-1}$

end do

(backward substitution:)

$x_n \leftarrow b_n/d_n$

for $i = n-1, \dots, 1$ **do**

$x_i \leftarrow (b_i - c_i x_{i+1})/d_i$

end do

output($x_1, x_2, \dots, x_n$)

We can check that the above method requires

$$5n - 4$$

number of multiplications and divisions. If we compare it with the number of operations of the Gaussian elimination, which is $n^3/3$ multiplications and divisions, then we can see that for a tridiagonal system this special algorithm should be applied.

We know that if the tridiagonal matrix $\mathbf{A}$ is also diagonally dominant, then the above algorithm can be performed (without pivoting).

3.5 Simultaneous Linear Systems

Frequently we would like to solve so-called **simultaneous linear systems**, i.e., systems of the form

$$\mathbf{A}\mathbf{x} = \mathbf{b}^{(i)}, \quad \text{for } i = 1, \dots, m.$$

We can shortly write the above system as

$$\mathbf{A}\mathbf{X} = \mathbf{B},$$

where the i th columns of the $n \times m$ dimensional matrix

$$\mathbf{B} = (\mathbf{b}^{(1)}, \mathbf{b}^{(2)}, \dots, \mathbf{b}^{(m)})$$

is $\mathbf{b}^{(i)}$, and the i th column of the $n \times m$ dimensional matrix

$$\mathbf{X} = (\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(m)})$$

is $\mathbf{x}^{(i)}$, i.e., the solution of the system

$$\mathbf{A}\mathbf{x}^{(i)} = \mathbf{b}^{(i)}.$$

Since pivoting in the Gaussian or Gauss–Jordan elimination depends only on the coefficient matrix, it can be performed on the $n \times (n + m)$ dimensional augmented matrix. For example, if we perform the Gauss–Jordan elimination on the augmented matrix

$$(\mathbf{A}, \mathbf{B})$$

we get a matrix of the form

$$(\mathbf{I}, \mathbf{X}).$$

Then the solution of the simultaneous linear system

$$\mathbf{X}$$

appears in the last m columns of the augmented matrix.

3.6 Matrix Inversion and Determinants

The inverse $\mathbf{A}^{-1}$ of a nonsingular square matrix $\mathbf{A}$ satisfies the matrix equation

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{I},$$

so the matrix inversion is equivalent to solving the simultaneous linear system

$$\mathbf{A}\mathbf{X} = \mathbf{I}.$$

We can use the Gauss–Jordan elimination to solve this problem. It can be checked that the number of operations needed to compute the matrix inverse with the Gauss–Jordan elimination is

$$\frac{3}{2}n^3 + \mathcal{O}(n^2)$$

number of multiplications and divisions.

Example

Compute the inverse of the matrix

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \\ -2 & 0 & -1 \end{pmatrix}.$$

We use the Gauss–Jordan elimination:

$$\begin{pmatrix} 1 & 0 & 2 & 1 & 0 & 0 \\ -1 & 1 & 0 & 0 & 1 & 0 \\ -2 & 0 & -1 & 0 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 1 & 0 \\ 0 & 0 & 3 & 2 & 0 & 1 \end{pmatrix} \sim$$

$$\begin{pmatrix} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 1 & 0 \\ 0 & 0 & 3 & 2 & 0 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & 0 & 0 & -\frac{1}{3} & 0 & -\frac{2}{3} \\ 0 & 1 & 0 & -\frac{1}{3} & 1 & -\frac{2}{3} \\ 0 & 0 & 3 & 2 & 0 & 1 \end{pmatrix} \sim$$

$$\begin{pmatrix} 1 & 0 & 0 & -\frac{1}{3} & 0 & -\frac{2}{3} \\ 0 & 1 & 0 & -\frac{1}{3} & 1 & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{2}{3} & 0 & \frac{1}{3} \end{pmatrix}$$

Example cont.

Hence

$$\mathbf{A}^{-1} = \frac{1}{3} \begin{pmatrix} -1 & 0 & -2 \\ -1 & 3 & -2 \\ 2 & 0 & 1 \end{pmatrix}.$$

Certainly, we can use pivoting techniques together with the Gauss-Jordan elimination for computing the inverse matrix if we would like to reduce the rounding errors or to avoid division by zero.

The Gaussian elimination with pivoting can be performed if and only if $\det(\mathbf{A}) \neq 0$. We have seen that

$$\det(\mathbf{A}) = (-1)^s \det(\mathbf{A}^{(n-1)}),$$

where s denotes the number of row changes. Therefore, the determinant is equal to the product of the pivot elements with an appropriate sign:

$$\det(\mathbf{A}) = (-1)^s a_{11}^{(1)} a_{22}^{(2)} \cdots a_{nn}^{(n-1)}.$$

Example

Consider the coefficient matrix of a previous example:

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & -2 & -2 \\ 2 & -1 & 2 & 4 \\ -1 & 2 & 3 & -4 \\ -2 & 1 & 4 & -2 \end{pmatrix}.$$

Compute the determinant of $\mathbf{A}$. We have performed the Gaussian elimination on $\mathbf{A}$ and got

$$\mathbf{A}^{(3)} = \begin{pmatrix} 1 & -2 & -2 & -2 \\ 0 & 3 & 6 & 8 \\ 0 & 0 & 1 & -6 \\ 0 & 0 & 0 & 38 \end{pmatrix}.$$

Therefore, $\det(\mathbf{A}) = \det(\mathbf{A}^{(3)}) = 1 \cdot 3 \cdot 1 \cdot 38 = 114$.